Order and Repetition

$n =$ 0:	1	1	1
$n =$ 1:			
$n =$ 2:			
$n =$ 3:			
$n =$ 4:			
$n =$ 5:			
$n =$ 6:			
$n =$ 7:			
$n =$ 8:			
$n =$ 9:			

First add diagonal lines and labels the diagonal 'columns' with k-values to the above figure to make your life easier. remember the initial diagonal column corresponds to $k = 0$.

Basic Building Formula: The way you construct the next line of Pascal's triangle is by adding adjacent entries and placing the sum in the line below. For example, the on the projection in green or in the table above that $\binom{2}{0} + \binom{2}{1} = \binom{3}{1}$. Can you generalize this to $\binom{3}{k-1} + \binom{3}{k} =$? Can you further generalize this to $\binom{n-1}{k-1} + \binom{n-1}{k} =$?

Left-Right Symmetry Formula: There is a left-right symmetry of Pascal's Triangle. For example, the on the projection in blue or in the table above that $\binom{7}{2} = \binom{7}{5}$. Can you generalize this to $\binom{7}{k} =$? Can you further generalize this to $\binom{n}{k} =$?

Hockey Stick Formula:Consider the following illustrated in yellow on the projection: $\binom{2}{2} + \binom{3}{2} + \binom{4}{2} = \binom{5}{3}$. Can you generalize this to $\binom{n}{n} + \binom{n+1}{n} + \binom{n+2}{n} =$? Can you generalize this to $\binom{n}{n} + \binom{n+1}{n} + \cdots + \binom{n+k}{n} =$?

Sum of Rows Formula: What is the sum of elements in a row of Pascal's triangle? Can you find a formula for this?

Alternating Sum of Rows Formula What is the alternating sum of elements in a row of Pascal's triangle? Alternating sum means the add the first, subtract the second, add the third, and so on? For example $\binom{4}{0} - \binom{4}{1} + \binom{4}{2} - \binom{4}{3} + \binom{4}{4} = ?$. What is a general formula for this for $n \geq 1$?